Transaction Operation for Certificate Asset Creation, Update, and Emitting event

Implementation of transaction for creating a BSC certificate

Input: BSCcertificateId, registrationNo, session, firstName, lastName, age, idNo, fathersName, mothersName, cgpa, department, universityName, country, date

```
BSCcertificate Registry ← Entity  of getAssetRegistry('org.BSC.assetTxn.BSCcertificate')
factory ← Instance of  getFactory()
NS ← Instance of org.BSC.assetTxn

BSCcertificate ← Entity  of factory.newResource(NS,'BSCcertificate',BSCdata.BSCcertificateId)

BSCcertificate.owner ← BSCdata.owner
BSCcertificate.registrationNo ← BSCdata.registrationNo
BSCcertificate.session ← BSCdata.session
BSCcertificate.firstName ← BSCdata.firstName
BSCcertificate.lastName ← BSCdata.lastName
BSCcertificate.age ← BSCdata.age
BSCcertificate.idNo ← BSCdata.idNo
BSCcertificate.fathersName ← BSCdata.fathersName
BSCcertificate.mothersName ← BSCdata.mothersName
BSCcertificate.cgpa ← BSCdata.cgpa
BSCcertificate.department ← BSCdata.department
BSCcertificate.universityName ← BSCdata.universityName
BSCcertificate.country ← BSCdata.country
BSCcertificate.date ← BSCdata.date

event ← Entity of factory.newEvent(NS,'BSCcertificateCreated')
    event.BSCcertificateId ← BSCdata.BSCcertificateId
    emit(event)
 return BSCcertificateRegistry.add(BSCcertificate)
```

Implementation of transaction for updating a BSC certificate

Input: BSCcertificateId, newsession, newfirstName, newlastName, newfathersName, newmothersName, newcgpa

```
BSCcertificate Registry ← Entity  of getAssetRegistry('org.BSC.assetTxn.BSCcertificate')
factory ← Instance of  getFactory()
NS ← Instance of org.BSC.assetTxn

BSCcertificate.BSCcertificateId ← updatesData.BSCcertificateId
BSCcertificate.session ← updatesData.newsession
BSCcertificate.firstName ← updatesData.newfirstName
BSCcertificate.lastName ← updatesData.newlastName
BSCcertificate.fathersName ← updatesData.newfathersName
BSCcertificate.mothersName ← updatesData.newmothersName
BSCcertificate.cgpa ← updatesData.newcgpa

 event ←  factory.newEvent(NS,'BSCcertificateUpdated')
    event.BSCcertificateId ← updatesData.BSCcertificateId
    emit(event)
   return bC.update(BSCcertificate)
```

Validating Certificate authentication by access transaction and access registers

```
Query AccessHistorianRecord:
        description ← "Returns Historian Records"
              statement ← SELECT org.hyperledger.composer.system.
                          HistorianRecord where (transactionType == _$input)
Query AccessAssetRegistry:
              description ← "Returns Asset Registry Elements
              statement ← SELECT org.hyperledger.composer.system.AssetRegistry
                          where(assetID ==$input)
acessSpecificAsset:
                    bnUtil ← Business Network Definition of Certificate System
              assReg ← bnUtil.connection.query('AccessAssetRegistry')
              qry ← bnUtil.connection.buildQuery(assReg)
              return bnUtil.connection.query(qry,{input: assReg#ID)
acessUpdateAssetTransaction:
                     bnUtil ← Business Network Definition of Certificate System
               histRecs ← bnUtil.connection.query('AccessHistorianRecord')
               qry← bnUtil.connection.buildQuery(assReg)
               return bnUtil.connection.query(qry,{input: 'updateCertificateInfo')
checkAssetValidity:
              Input: Asset/BSCcertificateID
              asset ← accessSpecificAsset(assReg#ID)
              if asset IS NOT void
                then historianRecs ← accessUpdateAssetTransaction()
                    for record in historianRecs:
                        if record.BSCcertificateID IS assReg#ID
                        return record
              else
                  return VOID
```

Participant Creation, Assigning Identities and Providing Participants the Access to Execute Transactions

Assigning Identities to End-Use Participant:

participantInstance←composer.participant.add(class:EndUser,participantkey:6789,fname: Shamma, lname: Dilshad, contactNo: 01718421092, email: shammadilshad@gmail.com)

composer.identity.issue(participantInstance,admin@certificate-securing-ledger)

Access Control for System Admin and End Users:

Rule AdminSystemResource:

description← "Grant business network administrators full access to system resources"
participant← "org.hyperledger.composer.system.NetworkAdmin"
operation← ALL
resource← "org.hyperledger.composer.system.**"
action← ALLOW

Rule EndUserPermissionHistorianRecordREAD_CREATE:

description← "Can VIEW and CREATE HistorianRecord inside the Historian"
participant← "org. certificate-securing-ledger.participants.EndUser"
operation← READ, CREATE
resource← "org.hyperledger.composer.system.HistorianRecord"
action← ALLOW

Rule EndUserPermissionAssetRegistryREAD:

description← "Can only VIEW Assets inside AssetRegistry"
participant← "org. certificate-securing-ledger.participants.EndUser"
operation← READ
resource← "org.hyperledger.composer.system.AssetRegistry"
action← ALLOW

Rule EndUserPermissionUpdateCertificateInfoCREATE:

description←"Can EXECUTE UpdateCertificiateInfo transaction"
participant← "org. certificate-securing-ledger.participants.EndUser"
operation← READ
resource← "org. certificate-securing-ledger.certificateInfo. UpdateCertificateInfo"
action← ALLOW

Serial No	ID	Student Name	CGPA	Dept.	Institution
01	1106028	Shamma Dilshad	3.85	CSE	BAIUST
02	1106026	Saima Ahmed	3.78	CSE	BAIUST
03	1205016	Ehsanul Haque	3.90	EEE	BAIUST
04	120529	Zerin Tasnim	3.80	CSE	BAIUST
05	1306013	Fatema Simi	3.70	BBA	BAIUST
06	1306016	Syed Resalat	3.65	BBA	BAIUST